

## Topic 2 of Personnel Economics: Efficiency and extensions

### Exercises

#### Exercise 2.1

Consider a firm which sells its products on the market for price  $p$ . Production depends on how much effort the firm's workers exert. For simplicity, assume that the firm employs only one worker. Production technology is described by:  $q = \phi e$ , where  $q$  is the firm's output, and  $e$  is the worker's effort. The worker's utility function is:

$U = w - \frac{1}{2}\theta e^2$ , where  $w$  is the worker's wage. The worker's participation constraint is  $U \geq V$ . The firm cannot observe effort, but only output. The firm offers a wage scheme:  $w = a + bq$ .

- a) Derive the profit-maximizing level of  $a$  and  $b$ , taking into account the worker's participation constraint and effort choice. Use backward induction and discuss your intermediate results along the way.
- b) Calculate the level of profits. Discuss under what conditions the level of profits is positive.

#### Exercise 2.2

Consider the same setting as in Exercise 2.1, but now assume that the worker likes to exert effort to some extent. This is reflected in his utility function, which is:

$U = w + \gamma e - \frac{1}{2}\theta e^2$ . The term  $\gamma e$  represents the joy of work. All other variables remain the same.

- a) Suppose the firm offers a fixed salary to its worker and no bonus pay ( $a > 0$ ,  $b = 0$ ). How hard does the worker work? What is the level of the fixed pay so that the firm attracts the worker at lowest cost? What is the level of profits? Interpret your results.
- b) Suppose that the firm offers a wage scheme:  $w = a + bq$ . Derive the profit-maximizing level of  $a$  and  $b$ , taking into account the worker's participation constraint and effort choice. Compare your results with those of 2.1a above. Provide interpretations, in particular for the effect of  $\gamma$  on the results.

#### Exercise 2.3

Consider the model of Exercise 2.1. Derive the social welfare maximizing level of effort. What level of the bonus  $b$  results in maximum social welfare? Interpret the result in non-technical words!

#### Exercise 2.4 (copied (slightly adapted) from Peter Kuhn's teaching material)

Suppose the production function in a particular principal-agent relationship is  $Q = 8E$  and the agent's utility function is  $U = Y - 2E^2$ . The price of the product is equal to one. Suppose that the agent's best alternative to this job gives a utility of  $U^{alt} = 4$ .

- a) Find the contract  $(a, b)$  that yields the highest possible level of profits, subject only to the agent's participation constraint,  $U \geq U^{alt}$ .
- b) Next, find the contract  $(a, b)$  that yields the highest possible level of worker utility, subject to the constraint that the firm earns non-negative profits ( $\Pi \geq 0$ ). Compare these two contracts. What do they have in common?

**Exercise 2.5** (this is a Midterm question of 2021)

Suppose workers can work on two tasks: task 1 and task 2. The variable  $g \geq 0$  denotes effort exerted on task 1 and the variable  $h \geq 0$  denotes effort exerted on task 2. The utility function of each worker is  $U = w + \gamma g - \frac{1}{2}\theta g^2 - \frac{1}{2}\theta h^2 - \frac{1}{2}\theta(g+h)^2$ , where  $w$  is the worker's income,  $\gamma > 0$  reflects the worker's intrinsic motivation and the final three terms reflect costs of effort. The worker's income consists of a base salary  $z$  plus a bonus of  $b$  for each unit of effort exerted on task 2. That is,  $w = z + bh$ . The worker aims for maximum utility. We assume throughout that  $0.5\gamma < b < 2\gamma$ .

- a) Derive the levels of effort the worker chooses to exert on task 1 and task 2.
- b) Explain in at most three sentences why a higher bonus for effort exerted on task 2 leads to a lower level of effort exerted on task 1.
- c) Give two possible reasons for why the employer only gives a bonus for effort exerted on task 2 and not for effort exerted on task 1.
- d) One way to deal with a multitask problem is to redesign jobs. Explain in at most three sentences how redesigning jobs can solve a possible multitask problem in the context of this exercise.
- e) Explain in one sentence why it can be useful to develop and analyze an economic model.

## Answers

### 2.1

a)  $b = p$  and  $a = V - \frac{1}{2} (p\phi)^2 / \Theta$ .

b) Profits equal  $= -V + \frac{1}{2} (p\phi)^2 / \Theta$ . They can only be positive if the worker's outside option is not too good, the price of the product is sufficiently high, the productivity of effort is sufficiently high and the worker's cost of effort is sufficiently low.

### 2.2

a) The first-order condition for optimal effort is  $\gamma - \Theta e = 0$ , and so  $e^* = \gamma / \Theta$ . Worker works harder, the higher is  $\gamma$ . Participation constraint:  $a + \gamma e - \frac{1}{2} \Theta e^2 = V$ . Substitute  $\gamma / \Theta$  for  $e$  and rewrite,  $a = V - \frac{1}{2} \gamma^2 / \Theta$ . So, worker gets a lower wage the more he likes his job (the higher  $\gamma$ ). Profits  $= p\phi \gamma / \Theta - (V - \frac{1}{2} \gamma^2 / \Theta)$ . Higher  $\gamma$  gives higher profits as effort is higher and wage is lower.

b)  $b = p$  and  $a = V - \frac{1}{2} (p\phi)^2 / \Theta - \frac{1}{2} \gamma^2 / \Theta - \phi p \gamma / \Theta$ . Optimal bonus is exactly the same as in the absence of joy of effort. Fixed pay is lower.

### 2.3

Social welfare is the sum of the principal's profit and the agent's utility:  $p\phi e - w + w - \frac{1}{2} \Theta e^2$ . Note that the two "w"-s cancel! Give that some thought! Differentiate to  $e$  gives  $p\phi - \Theta e = 0$ . Interpret all terms of this first-order condition! Rewriting gives  $e = [p\phi] / \Theta$ . In Exercise 2.1 we already derived that the worker chooses  $e = [b\phi] / \Theta$ . Clearly,  $b=p$  results in maximum social welfare!

### 2.4 *(copied from Peter Kuhn's teaching material)*

a) Maximizing profits subject to the participation constraint of the worker yields  $b = 1$  and  $a = -4$ .

b) Maximizing worker's utility subject to the participation constraint of the firm yields  $b = 1$  and  $a = 0$ .

Both have a 100% commission rate to maximize the size of the pie (i.e. to achieve efficiency). Within the set of efficient contracts that is acceptable to both parties, the parameter  $a$  can then be adjusted to achieve any desired distributional outcome.

### 2.5

a) The worker maximizes utility to  $g$  and  $h$  resulting in the first-order conditions:

$$\gamma - \theta g - \theta(g+h) = 0$$

$$b - \theta h - \theta(g+h) = 0$$

Before solving, take a minute to think about the meaning of these marginal benefits and costs. Next, solving gives  $g = (2\gamma - b)/3\theta$  and  $h = (2b - \gamma)/3\theta$ . (Note: the two first-order conditions are two equations and two unknowns and so can be solved. If you forgot how, check your math course of last year.)

b) A higher bonus for effort exerted on task 2 makes it optimal for the worker to increase effort on task 2 as it yields higher monetary returns. This, in turn, increases the marginal cost of effort of task 1. Hence, the worker reduces the effort exerted on task 1.

It is interesting to note that, if the principal cares about output  $g$ , the multitasking problem makes paying a high bonus less attractive to the principal. By raising the bonus, output  $h$  increases but output  $g$  decreases. As a result, it will generally be optimal for the principal to pay a lower bonus than in the absence of a multitasking problem. That is, multitasking is one reason for principals not to pay a bonus equal to the price of the product that is produced. If the principal cares a lot about output  $g$ , the principal may even completely refrain from paying a bonus, i.e. set bonus  $b$  equal to 0.

c) Effort on task 1 may be unobservable to the employer and hence noncontractible. Or the employer may simply not care about the effort on task 1.

d) Create jobs where workers only perform task 2 and create jobs where workers only perform task 1. For the latter job, try to find workers who are highly intrinsically motivated (i.e., have high  $\gamma$ ). The workers in the jobs where only task 2 is performed can be incentivized to work hard without causing a multitask problem.

e) It can be helpful in checking the consistency of one's arguments. And while analyzing, it can bring new unexpected insights.